

State and local economic impacts from wind energy projects: Texas case study

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ARTICLE INFO

Article history:

Received 9 May 2011

Accepted 23 September 2011

Keywords:

Wind energy

Texas

Economic impacts

ABSTRACT

This paper uses the Jobs and Economic Development Impacts (JEDI) model to estimate economic impacts from 1398 MW of wind power development in four counties in west Texas. Project-specific impacts are estimated at the local level (i.e., within a 100-mile radius around the wind farms) and at the state level. The primary economic policy question addressed is how investment in wind energy affects the state and local communities where the wind farms are built. During the four-year construction phase approximately 4100 FTE (full time equivalents) jobs were supported with turbine and supply chain impacts accounting for 58% of all jobs generated. Total lifetime economic activity to the state from the projects equated to more than \$1.8 billion, or \$1.3 million per MW of installed capacity. The total economic activity to the local communities was also substantial, equating to nearly \$730 million over the assumed 20-year life cycle of the farms, or \$0.52 million per MW of installed capacity. Given the current level of impacts observed, and the potential for increased impacts via greater utilization of in-state manufacturing capacity and the development of trained wind industry specific laborers, Texas appears to be well positioned to see increasing impacts from continued wind development.

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1. Introduction

Wind power is recognized as an important energy resource throughout the world. In several scenarios of future electricity costs, wind power is forecast to produce electricity at lower costs than that of biomass and solar-pv for decades (e.g., EREC, 2008; de Vries et al., 2007). Wind energy has also been recognized as one of the most environmentally benign sources of electricity generation: it causes no emissions of harmful pollutants, including the greenhouse gas carbon dioxide; does not require mining or drilling for fuel; does not cause radioactive or hazardous wastes; and does not use water for steam generation or cooling (Brittan, 2002; Schiermeier et al., 2008; Warren et al., 2005). Because the environmental benefits of wind energy appear significant, public support for expanding wind energy development is often high (Swofford and Slattery, 2010).

Social and political support for wind energy has turned it into one of the fastest growing sources of power generation in the world.¹ Growth in the U.S. achieved 25–50% per year between 2005 and 2009 (Fig. 1), and wind energy now supplies approximately 2.5% of annual

U.S. electricity consumption (Wiser and Bolinger, 2010). Interconnection queues around the U.S. suggest that planning for new wind development continues at a strong pace (Wiser and Bolinger, 2010).² However, the dramatic growth in wind power development has raised a number of challenges for the industry. For wind energy to provide 20% of U.S. electricity needs by 2030, an energy scenario modeled by the U.S. DOE (DOE, 2008), challenges in the areas of technology, manufacturing, and transmission and grid operations must be overcome. Moreover, existing federal policies expire in 2012 creating federal policy uncertainty that has the potential to affect sustained growth rates observed in the recent past. Additional industry challenges include concerns over potential impacts to wildlife, particularly birds and bats (Arnett et al., 2008; Kunz et al., 2007), visual and noise impacts on communities (Devine-Wright, 2005a, b; Johansson and Laike, 2007; Pedersen and Wye, 2007; Pedersen et al., 2009; Swofford and Slattery, 2010), and the ability to more clearly quantify the system-wide environmental and emissions impacts from wind energy (Sims et al., 2003).

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¹ Recent growth has resulted from concern over climate change, energy security, the cost of fossil fuels, as well as from policy support which has made wind a viable investment opportunity in specific markets (Wiser and Bolinger, 2010).

² Capacity queuing for interconnection in regional transmission operator (RTO), independent system operator (ISO), and utility systems is on the order of 300 GW, far exceeding interconnection requests for all other power generation resources in the U.S. While it is likely that only a fraction of these projects will be built this demonstrates that interest in continued wind development is substantial (Wiser and Bolinger, 2010).

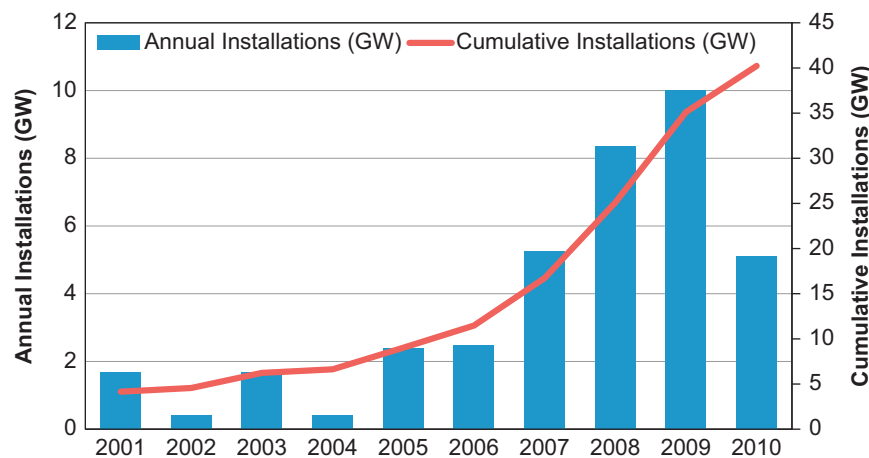


Fig. 1. Annual and cumulative installations of wind energy in the U.S. Expirations of the federal production tax credit in 2001 and 2003 are evident in the decline in annual installations the following year (Source: AWEA, 2010).

Many of the impacts from wind energy projects are recognized as predominately local (e.g., wildlife displacement, visual and noise impacts, economic development impacts). Thus, while general public and political support for wind energy is high, siting wind plants frequently raises concerns in local communities.³ Among these concerns are questions about the economic impacts of wind power development especially where large wind farms are built near small, rural communities. Wind energy advocates often argue that local economic development impacts are a notable benefit for host communities, and local economic impacts are a fundamental concern for local elected officials who may be primarily interested in tax revenue and employment impacts.⁴ Critics on the other hand have asserted that projects have little lasting local economic impact. In today's economic climate, where job creation is paramount, such questions take on a greater degree of significance. In the past, studies of economic impacts from building and operating wind plants suggest that economic impacts to states are substantial (e.g., Lantz, 2008; Lantz and Tegen, 2011; Pedden, 2006; Tegen, 2006; Reategui and Tegen, 2008). However, few studies have sought to quantify these impacts at the county or local level, particularly where wind power plants are built in rural areas with few supporting industries.

This paper uses the National Renewable Energy Laboratory's (NREL's) Jobs and Economic Development Impacts (JEDI) model to assess economic impacts from wind power development in four counties in west Texas. Project-specific impacts are estimated at the local (i.e., within a 100-mile radius around the project) and at the state level. In addition, modeling is completed both with industry supplied project-specific data and default JEDI inputs. The former approach allows a direct comparison between state and local economic impacts from these projects while the latter allows a test of the validity of the NREL JEDI Wind model for the purpose of informing its future development. This analysis focuses on Texas—the leader in wind power development in the U.S. with over 10,085 MW of installed capacity⁵ (AWEA, 2010). In 2009 alone, energy companies invested more than \$3 billion to install almost 2300 MW of capacity in the state. This investment in Texas increased total U.S. wind capacity by roughly 7%. The primary economic policy question addressed in this study is how

this investment affects the state and local communities where the wind farms are built. Results presented here include economic development impacts such as job creation, salaries and benefits, and impacts on overall economic output (activity) in the communities where the wind projects are located. This analysis also discusses emerging trends that are likely to influence economic impacts of continued wind energy deployment in Texas.

1.1. Why local impacts analysis is important

The current economy has placed a significant degree of attention on economic and fiscal policy at the federal level; however, many rural communities have faced economic challenges long before the most recent recession. The development of new wind generation creates the opportunity for construction, manufacturing, maintenance and other jobs, increases tax revenues, and provides lease income for landowners. Moreover, wind plants are built where wind resources are highest, typically in rural areas and far from electrical loads. Wind energy creates the opportunity for large new investments in rural communities across the U.S.

However, economic impacts are multi-faceted and variable. Project level impacts depend on available resources and the ability of local businesses to participate in wind energy projects as well as the preferences of individual contractors (Lantz and Tegen, 2008). The extent of local ownership along with the sourcing of large capital items (e.g., blades and towers) can also dramatically impact the scale of economic impacts experienced by a state versus a local community (Lantz and Tegen, 2008; 2009). In the most extreme cases, wind plants can be built with little economic value for the locality or host community where a project is sited. Tracing the distribution of impacts within the state and local economy is fundamental to understanding the value of wind energy projects for the localities where projects are cited; it can also help to inform new policy development to ensure that states and localities are capturing the impacts they desire.

1.2. Literature review

Historically, wind energy economic impacts analysis in the U.S. has emphasized impacts to the state or national economy (e.g., DOE, 2008; Lantz and Tegen, 2008; Reategui and Tegen, 2008). Past research has also sought to make comparisons between wind energy technology and alternative power generation technologies (e.g., Tegen, 2006; Lantz and Tegen, 2008). Additional, research has been conducted to understand the impacts of local ownership

³ Concerns are often raised over perceived injustices in the allocation of project costs and benefits (i.e., host communities are forced to bear the majority of social costs but only receive a minority of project benefits).

⁴ Other local stakeholders may also be concerned with the visual, landscape, or noise impacts of wind plants located in their communities.

⁵ This would rank Texas sixth in the world in wind energy capacity if it were a country, behind Germany, the rest of the U.S., Spain, China, and India.

(e.g., Costanti, 2004; DanMar & Associates, 1996; GAO, 2004; Lantz and Tegen, 2009; Kildegaard and Myers-Kuykindall, 2006).

The economic impact of wind energy projects to states varies although a typical 100 MW wind power plant will require a total investment of approximately \$200 million dollars (Lantz and Tegen, 2008). In terms of jobs, the same 100 MW wind plant is likely to support an average of 80–100 construction workers onsite for a period of one year during construction and 6–8 onsite operations and maintenance (O&M) workers annually throughout the life of the plant (NREL Database).⁶ Wind projects provide property tax payments to local government on the order of \$4000–\$12,000 per MW and landowner lease payments on the order of \$3000–\$7000 per MW (NREL Database).

Wind project economic impacts, however, are not limited to onsite or direct impacts. Work onsite frequently generates increased business activity for the local hospitality industry, including hotels and food vendors. Additionally, projects may rely on local quarries and concrete for the raw materials utilized in project roads and foundations. Finally, because a majority of project costs are in the purchasing of hardware and equipment (turbines alone are approximately 70–75% of total project cost), much of the economic activity and job support potential accrues to manufacturers and the broader industry supply chain. States and localities with the capacity to provide major turbine components (e.g., towers, blades, nacelles) stand to see their construction period economic output from wind energy investments increase by more than a factor of three when they increase the share of turbines supplied by in-state manufacturers from 0% to 50% (Lantz and Tegen, 2008). A local ownership component in wind projects has also been shown to increase construction and operation period jobs impacts by a factor of 1–3 times (Lantz and Tegen, 2009).

Analysis of gross impacts to the national economy suggests impacts on the order of hundreds of thousands of jobs (DOE, 2008). However, the net impact on the national economy is less clear and no peer-reviewed analysis of net economic impacts from renewable energy deployment in the U.S. has been conducted to date. Research from the international context suggests that the net economic impact (including net job creation) from deploying renewable energy technology in the European Union is likely to be positive but not substantial (Ragwitz et al., 2009).⁷ Net economic impacts are generally believed to be a function of the long-term cost of renewable energy technologies relative to conventional technologies (and their fuels) as well as expectations regarding the ability to manufacture and export technology (Lehr et al., 2008). Because of these variables, the long-term net impact to the U.S. economy from renewable energy deployment has a high level of uncertainty.

2. Methodology

2.1. Model structure and study sites

To evaluate the economic impacts of wind energy projects in Texas, this study utilized the JEDI Wind model.⁸ The appropriateness

of this model for this study was determined by its prior use by the U.S. Department of Energy (US DOE), U.S. Department of Agriculture (USDA), the National Renewable Energy Laboratory, as well as several universities, all of whom have used the JEDI model to assess the economic impacts of wind plants (e.g., DOE, 2008; GAO, 2004; Lantz and Tegen, 2008, 2009; Loomis and Hinman, 2010; Torgerson et al., 2006).

The JEDI Wind model is a spreadsheet tool that applies standard input–output (I/O) multipliers and consumption patterns; multiplier data are provided by the Minnesota IMPLAN Group (MIG).⁹ Analysis with the JEDI tool can be carried out at the county, state, regional, or national level. In line with the basic I/O methodology, economic development impacts from wind energy projects are estimated based on the relationship between past expenditures and the resulting economic activity, in those sectors affected by wind energy projects.¹⁰ Industry-specific multipliers used in this analysis are from 2007 and reflect industry relationships in Texas and the localities of interest at that time.¹¹

Basic JEDI Wind user inputs include high level project information (i.e., year of construction, project size, turbine size, installed cost, and O&M costs). However, the JEDI tool also allows the user the ability to adjust or modify the allocation of project costs between specific industries affected by a project (during development, construction, and operations) and the percentage of project expenditures (within a specific industry) that go to businesses and contractors based in the area under consideration for development (i.e., the local purchase coefficient).

The JEDI Wind model reports gross economic impacts in the form of jobs (full time equivalents (FTE) for a single year), earnings (wages, salaries, and associated benefits), and economic output (a general measure of economic activity representing the total value of production). As a gross impacts model, JEDI Wind does not consider potential impacts on electricity prices or potential worker displacement from other sectors; it also assumes projects can be economically justified on their own merits (i.e., as a profitable enterprise, capable of attracting sufficient financing to be built). Model results are categorized as impacts occurring during (1) development and construction and (2) O&M. Further, each of these categories consists of three tiers; direct impacts (onsite), indirect impacts (supply chain), and induced impacts, as shown in Table 1 (construction phase impacts) and Table 2 (operations period impacts).

In this study, the JEDI Wind model is used to evaluate economic impacts from two wind energy projects at both the state and local level. One of the projects considered was the Capricorn Ridge Wind Farm, a 407-turbine facility with 662.5 MW of installed capacity completed in 2008. The second project considered was the Horse

(footnote continued)

by Marshall Goldberg of MRG & Associates for NREL. It is a free tool and available for download. For more information on the full suite of JEDI models or to download the JEDI Wind model see: http://www.nrel.gov/analysis/jedi/about_jedi_wind.html.

⁹ The multipliers used in the JEDI models are derived using MIG economic data and IMPLAN Software. Original data sources, compiled by MIG, include: the U.S. Bureau of Economic Analysis, the U.S. Department of Labor, and the U.S. Census among others. For more information on IMPLAN multipliers, raw data, or software see: <http://implan.com>.

¹⁰ The relationship between expenditures (or demand) and the resulting economic activity is captured by the MIG multipliers. MIG multipliers are derived from analysis of historical demand (i.e., expenditures) and historical economic activity (i.e., jobs, earnings, output, etc.) in specific sectors of the economy, for a period of a single year. By determining the precise level economic activity resulting from past expenditures, this relationship or multiplier can then be used to project or approximate the economic activity resulting from new expenditures or proposed projects.

¹¹ Construction of the projects analyzed here occurred over the four-year time period from 2005 to 2008. 2007 multipliers are assumed to be generally representative of this time period.

⁶ The construction period generally determines the actual number of workers employed; however, in terms of man-hours a 100 MW project typically supports the equivalent of 80–100 full-time workers for a period of one year. Construction workers may or may not be from the state where a project is located. O&M employees more frequently reside in the state and often the community where projects are sited.

⁷ Impacts from an aggressive renewables scenario indicate 410,000 additional jobs and a 0.24% increase in gross domestic product would result by 2020 (Ragwitz et al., 2009).

⁸ The JEDI Wind model is a publicly available model used to estimate economic impacts of utility-scale wind energy projects. The model was developed

Table 1
Construction period impacts reported by JEDI Wind.

Category label	Description	Types of persons and businesses impacted
Project development and onsite labor impacts	Impacts from money spent on labor for persons working to develop and construct wind projects	Project managers, environmental technicians, civil engineers, legal staff, road builders, concrete pourers, crane operators, etc.
Local revenue, turbine, and supply chain impacts	Impacts resulting from equipment and turbine purchases. It includes impacts to OEMs and the array of suppliers providing components or other products for required wind plant equipment. It also includes impacts to the finance and banking sectors	Turbine, blade, and tower manufacturers, gear manufacturers, fiberglass and epoxy producers, steel producers, quarries, accountants, etc.
Induced impacts	Impacts from reinvestment and spending by beneficiaries of spending and economic activity in the top two tiers of impacts	Local retailers, food and hospitality services, childcare providers, etc.

Table 2
Operations period impacts reported by JEDI Wind.

Category label	Description	Types of persons and businesses impacted
Onsite labor impacts	Impacts resulting from money spent on labor for persons working to operate, maintain, and manage ongoing plant operations	Maintenance technicians, administrative staff and managers, etc.
Local revenue and supply chain impacts	Impacts from expenditures related to maintenance, repair, and general operation activities. Also includes impacts from land lease payments, property tax payments, insurance costs, and other ongoing expenses	Repair and replacement parts manufacturers, tool providers, local government, local utilities, insurance providers, welders and metal fabricators, etc.
Induced impacts	Impacts from reinvestment and spending by beneficiaries of spending and economic activity in the top two tiers of impacts	Local retailers, restaurants, childcare providers, etc.

Table 3
County and wind farm data.

Wind farm	County	Date of first construction	Date operational	Total MW	Total turbines	Combined county population ^a (% employable ^b)
Capricorn Ridge	Sterling and Coke	4th Q 2006	4th Q 2008	662.5	407	4605 (53.5)
Horse Hollow	Nolan and Taylor	2nd Q 2005	4th Q 2005	735.5	421	142,357 (65.1)

^a Data from 2005–2009 American Community Survey 5-Year Estimates (<http://www.factfinder.census.gov>).

^b Persons older than 16 years of age.

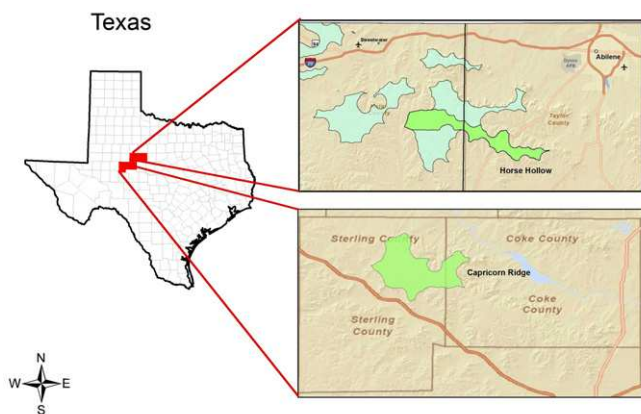


Fig. 2. Study area showing location of the two wind farms (green). Blue indicates other wind farms in the studies counties. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Hollow Wind Energy Center, a 421-turbine facility with 735.5 MW of installed capacity completed in 2006 (see Table 3, Fig. 2). Both of these projects have large footprints covering two adjacent counties.

Capricorn Ridge is located in extremely rural Coke and Sterling counties; Horse Hollow is sited in the more populous Nolan and Taylor two county area. Both projects are in west Texas.¹² Local multiplier data used in this analysis apply to each of the two-county areas where the projects are located.

In addition to analyzing economic impacts from these projects at the state and local level, default JEDI inputs were also used and then compared to the project-specific JEDI inputs. JEDI default inputs are generally representative of aggregate national wind industry averages and are derived from interviews with project developers, local government officials, utilities, and others in the power generation sector. Project-specific cost and state and local purchase coefficient data were primarily developed from project-specific data and records provided by the project owner. These data were supplemented by interviews with Texas state and local government officials and other industry representatives. Where project-specific data were not available default JEDI data were used. The local purchase coefficients applied in the

¹² Both county areas covered in this analysis contain other wind farms in addition to those studied here: Nolan and Taylor counties contain at least 10 additional wind projects.

Table 4
JEDI estimates of the economic activity to Texas.

	Total jobs		Earnings (\$M)		Output (\$M)	
	HH	CR	HH	CR	HH	CR
<i>Construction (annual average for 4 years)</i>						
Project development and onsite labor	68	102	\$4.8	\$7.0	\$5.7	\$7.8
Turbine and supply chain	285	308	\$15.5	\$17.1	\$50.7	\$58.0
Induced	122	140	\$5.7	\$6.5	\$17.4	\$19.9
Total	474	550	\$25.9	\$30.6	\$73.7	\$85.6
<i>Operations (annual for 20 years)</i>						
Onsite labor	33	30	\$1.9 ^a	\$1.7 ^a	\$1.9 ^a	\$1.7 ^a
Local revenue and supply chain	81	74	\$4.2	\$3.8	\$19.3	\$18.9
Induced	65	66	\$3.0	\$3.1	\$9.3	\$9.4
Total	180	170	\$9.2	\$8.6	\$30.6	\$30.0

^a Direct earnings and output are the same because the way the current model is structured, salaries and earnings from onsite labor are the only form of direct output.

analysis of local impacts are based on a 100-mile radius around each project.¹³

To demonstrate the incremental impact that would have resulted if Texas-based manufacturing or higher levels of Texas-based construction labor had been incorporated into these projects, two sensitivity scenarios are also included. In both these scenarios, project specific models were run after adjusting the inputs to independently consider 25% Texas manufacturing and 80% Texas construction labor.

2.2. Other modeling considerations

The JEDI model does not incorporate any spillover effect of local wind farms upon areas outside of the specific study area; it only captures those impacts that occur in the locality of interest. It also does not directly consider the impact of neighboring wind farms on the areas examined in this study.¹⁴ In addition, this analysis does not include impacts associated with spending of plant “profits” and assumes projects receive sales tax and property tax abatements. A portion of the property tax abatements are often replaced by payments in lieu of tax, which are included in these analyses. Finally, JEDI assumes a homogenous level of impact across the entire study region. In reality, impacts are likely to be concentrated in the population centers of rural counties.

The JEDI model is intended to construct a reasonable profile of investments (e.g., wind power plant construction and operating costs) and likely economic impacts resulting from the construction and operating periods. Given the potential for future changes in wind power plant costs and potential changes in industry and personal consumption patterns in the economy, the analysis is not intended to provide a precise forecast indicative of future projects, but rather an estimate of overall economic impacts from these specific projects.

3. Results

Economic impact results are reported for full time equivalent jobs, earnings, and economic output. Results are distinguished between the construction period, representing a short-term

infusion of investment and economic activity, and the operations period, representing a more modest but longer-term infusion of dollars into the local and Texas economy. For this study, the two sites were constructed over a four-year period. Therefore, construction period results are the average annual impact over the four-year construction period required to complete both projects. Operations period impacts represent the annual impact from ongoing project O&M. Because wind energy projects are typically financed on a 20-year basis, operations period impacts are expected to persist for at least 20 years.

3.1. Economic impacts to Texas

The estimated annual impact to Texas from building and operating these two wind projects is shown in Table 4. Total employment impacts to the state from each project are detailed in Fig. 3. The distribution of economic output over time is shown in Fig. 4.

During the four-year construction phase approximately 4100 FTE jobs were estimated to have been supported (1900 for Horse Hollow and 2200 for Capricorn Ridge), or approximately 1000 jobs per annum. Project development and onsite labor impacts accounted for 680 of these jobs (or 170 per annum) with turbine and supply chain impacts accounting for 58% of all jobs generated as a result of construction of the wind farms. The number of jobs supported at Capricorn Ridge was higher than at Horse Hollow (102 per annum versus 68 per annum). This result is primarily a function of higher construction costs (a general trend observed in the U.S. wind industry over this time period) and a moderately higher local purchase coefficient for this project resulting from increased use of Texas-based materials, goods, and services. During the O&M phase, an estimated 350 annual jobs are supported with no significant difference between the farms for onsite labor impacts, local revenue, supply chain impacts, and induced impacts. At Horse Hollow, 33 permanent onsite annual jobs are supported; at Capricorn Ridge, 30 permanent annual jobs are supported.

The jobs supported by these two wind farms were estimated to have generated \$57 million in earnings and \$160 million in economic output for the state per year during the four-year construction period. During O&M period, the statewide jobs were estimated to generate \$18 million per year in earnings and \$61 million per year in economic output. The 63 people hired onsite to permanently manage and operate the facilities generate approximately \$3.6 million in earnings annually equivalent to an average salary (and benefits) of \$58,000 per job.

Total lifetime economic activity to the state from the two wind farms is estimated at more than \$1.8 billion, assuming four years

¹³ Local purchase coefficient data of higher resolution, which would have allowed this analysis to focus the local results exclusively on the two county area represented by the I/O multipliers, were not available.

¹⁴ Indirectly effects from other projects may be captured if their presence leads to a greater availability of local goods and services, in effect additional wind projects in close proximity may increase the impact from any single project by concentrating labor and material resources in a specific area and reducing leakage of spending into the surrounding state(s).

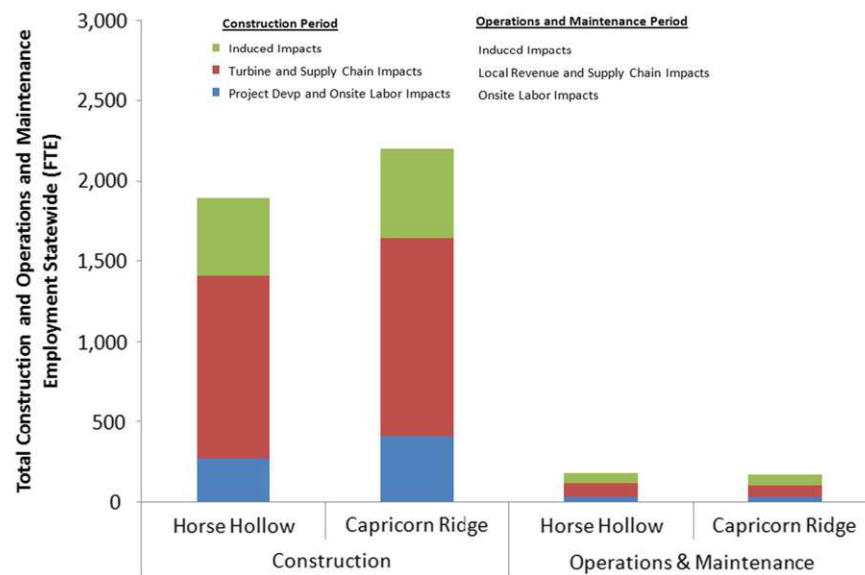


Fig. 3. JEDI results showing jobs supported statewide during construction and O&M (annual) at the studied wind farms.

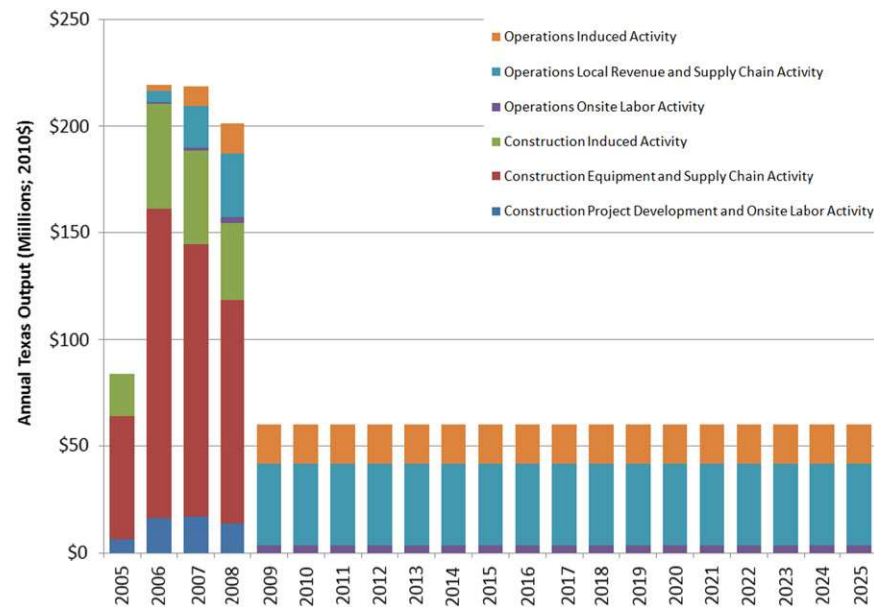


Fig. 4. JEDI results showing total economic output statewide during construction and operations and maintenance at the studied wind farms.

of construction and 20 years of operation (see Fig. 4). This equates to more than \$1.3 million per MW of installed capacity.

3.2. Economic impact to local communities

The estimated impact to those communities within a 100-mile radius of the farms, as computed by the JEDI model using the county-level multipliers, are detailed in Table 5. Total employment impacts to the local communities are detailed in Fig. 5. The distribution of economic output over time is shown in Fig. 6.

A total of approximately 900 FTE construction jobs within a 100-mile radius of these projects were supported by these projects (450 at Horse Hollow and 450 at Capricorn Ridge). On average these projects supported 225 workers per year for the four years of construction. Of the 225 FTE jobs supported, project development and onsite labor impacts accounted for 46 jobs (or 20%), while turbine and supply chain impacts accounted for 628

Table 5

JEDI estimates of the total economic activity to the local community (i.e., within 100 miles of the farms).

	Total jobs		Earnings (\$M)		Output (\$M)	
	HH	CR	HH	CR	HH	CR
<i>Construction (annual average for 4 years)</i>						
Project development and onsite labor	17	29	\$1.1	\$1.9	\$1.2	\$1.9
Turbine and supply chain	77	80	\$3.2	\$2.4	\$10.6	\$11.4
Induced	18	3	\$0.7	\$0.1	\$1.9	\$0.3
Total	113	112	\$5.0	\$4.3	\$13.8	\$13.6
<i>Operations (annual for 20 years)</i>						
Onsite labor	33	30	\$1.9	\$1.7	\$1.9	\$1.7
Local revenue and supply chain	65	60	\$2.5	\$1.4	\$12.5	\$11.2
Induced	30	7	\$1.1	\$0.2	\$3.2	\$0.7
Total	128	97	\$5.5	\$3.3	\$17.6	\$13.5

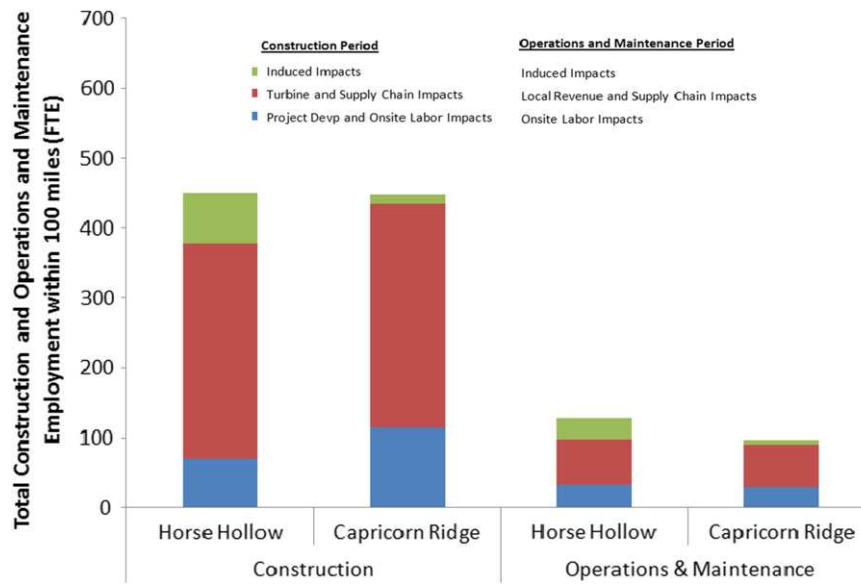


Fig. 5. JEDI results showing jobs supported within a 100-mile radius at the studied wind farms. Note that during O&M jobs are reported per annum.

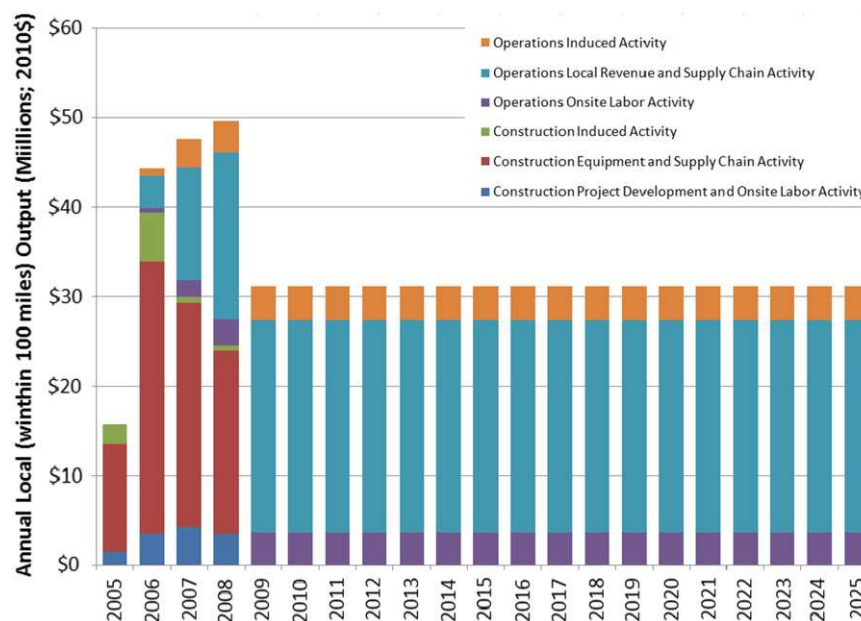


Fig. 6. JEDI results showing total economic output within a 100-mile radius of the studied wind farms.

jobs (or 70%). During the O&M phase, 225 local jobs are supported with the 63 onsite labor jobs the same number estimated using the statewide multipliers (see Table 4). This indicates that all onsite O&M jobs are filled by people living within 100 miles of the farms. The data also indicate that during the construction period, 22% of the Texas jobs supported by these projects were filled by workers living within 100 miles of the projects. During the O&M period, however, 64% of the Texas jobs were estimated to have been filled by workers coming from within this 100 mile zone.

The jobs supported locally by these two wind farms generated \$9.3 million in earnings and \$27 million in economic output for the local community per year during the four-year construction period (Table 5). During the O&M period, the jobs supported locally generated \$8.8 million per year in wages and \$31 million per year in economic output. The total economic activity to the local communities from the two wind farms was substantial,

equating to nearly \$730 million over the construction period and the assumed 20-year operating life of the farms, or \$0.52 million per MW of installed capacity (see Fig. 6).

3.3. Comparing JEDI default results and project-specific results

JEDI Wind default model inputs represent data aggregated from across the industry. However, input–output modeling tends to be sensitive to the modeling inputs, so best practice is to utilize as much project-specific data as possible. To illustrate the value of using project-specific data and to inform continued development of the JEDI Wind model, comparisons of model results using default inputs and project-specific inputs were also conducted. The default model inputs included only the installed MW capacity for each project and the national average project cost for the respective year of construction.

At the state level, the JEDI default model appears to overestimate economic impacts during construction and slightly underestimate impacts during operations relative to modeling conducted with the project-specific data (Fig. 7a). During construction the differences are largely explained by the fact that the project-specific installed costs were lower than the average nationwide installed costs for projects of this same vintage. This may be a result of some economies of scale garnered through such large projects and/or simple variability in equipment and labor costs based on the Texas market. During operations, the differences are likely the result of minor differences in O&M costs and direct payments (e.g., property tax, landowner lease payments) relative to state or national averages and the general accessibility of repair parts, equipment, and tools that is associated with a sizable market such as Texas.

At the local level, the JEDI default model appears to substantially underestimate construction period impacts and continues to underestimate impacts during operations relative to the results modeled using project-specific data (Fig. 7b). During construction the significant difference is the result of rather conservative local purchase coefficient assumptions applied for low population rural areas in the default model. In this specific case, there was much greater participation in the construction of these projects among the local communities than is assumed in the default model inputs. Greater construction period participation from local communities may result from the fact that project-specific data acquired for this analysis are based on a 100-mile radius around each project. Such a large area provides a larger resource base to draw from relative to a many rural counties. In addition, these regions of Texas have seen extensive wind energy development. It is possible that businesses have relocated to these areas to better serve the wind industry, or that existing local businesses have recognized wind energy as a significant opportunity and taken steps to actively participate in the industry. During operations the differences between the default model results and the project-specific results at the local level are more a function differences in O&M costs and/or modest differences in direct payments to

counties relative to what is estimated with the JEDI default inputs.

3.4. State level sensitivities on critical variables

The state level results highlighted above focus on the results of projects completed 3–5 years ago. However, the wind industry has grown and matured significantly over this time. In order to evaluate how the economic development impacts might look were these projects in development today two sensitivities, one including modest levels of Texas manufactured turbines (i.e., 25%) and one including an incremental increase to 80% Texas-based construction labor, are shown here. The consideration of Texas-based manufacturing is justified by the significant manufacturing investment in the U.S. that has occurred since the mid-2000s (Wiser and Bolinger, 2010). As the largest market for wind energy in the U.S. Texas has successfully attracted more than 15 wind energy manufacturers (Wiser and Bolinger, 2010) and, accordingly, the potential exists for projects today to utilize Texas-based manufacturing for wind turbine equipment (e.g., blades, towers, gearboxes, generators, etc.) and related electrical infrastructure (e.g., transformers, high voltage transmission lines, transmission towers, etc.). The possibility for increased use of Texas-based construction labor is justified by the emergence of various workforce development and training programs that equip the Texas labor pool for wind industry jobs in the state. Fig. 8 details the theoretical impact of increased reliance on Texas-based manufacturing for wind turbine and associated electrical infrastructure equipment for the projects considered here. Fig. 9 illustrates the theoretical impact to Texas from greater reliance on Texas-based construction labor for these projects. In the former case, construction period economic output is estimated to increase by nearly 140% if Texas manufacturing facilities had been able to supply 25% of turbine content (by cost) for these projects. Increasing use of Texas-based construction labor by 33% is estimated to have increased overall construction period employment (FTE) by approximately 8%.

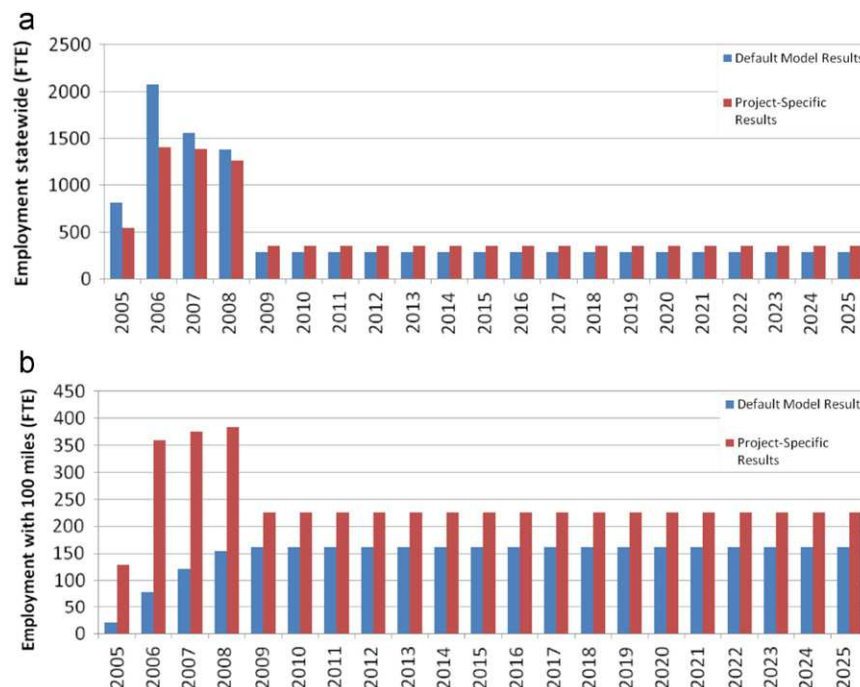


Fig. 7. JEDI results showing (a) number of supported jobs statewide and (b) number of supported jobs within a 100-mile radius of the studied wind farms estimated using default versus project-specific inputs.

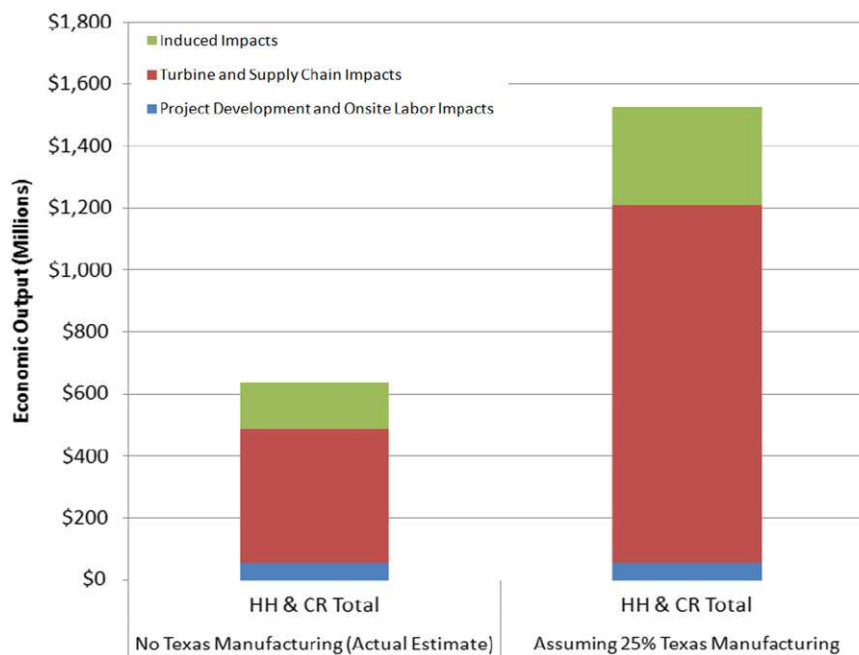


Fig. 8. Impact of increased reliance on Texas-based manufacturing for wind turbine and associated electrical infrastructure equipment for the projects considered.

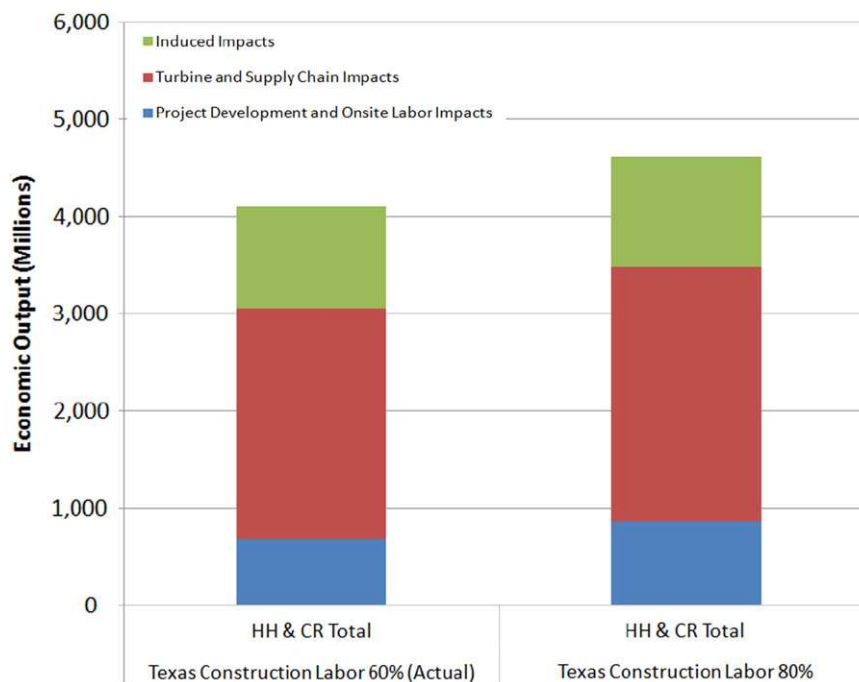


Fig. 9. Impact to Texas from greater reliance on Texas-based construction labor for these projects considered here.

4. Discussion

Economic development considerations in studies of wind farm impacts typically focus a great deal of attention on state and local jobs. Because of the relatively high capital costs for wind power projects, construction period jobs are an important piece of economic development impacts, especially at the local (100-mile radius) and county levels. Construction of a single wind power plant can present a short-term, significant, infusion of money into local communities. Indeed, previous JEDI research has shown that a state can supply more than 70% of the construction period jobs

(Lantz and Tegen, 2008). However, this category varies widely among projects and among developers, particularly in rural regions where construction labor may not be familiar with wind projects or may not be readily available. Local communities in this analysis received roughly 15–20% of overall construction period project-specific labor expenditures, indicating that these rural communities were able to capture a portion of the significant short-term construction investment. However, had the definition of local been limited to an area smaller than a 100-mile radius, a county, for example, the percent of capture may have been lower than 15%.

The results of this modeling work are broadly consistent with a recent review of 13 studies on the economic impact of wind farms in rural communities in the U.S. (Pedden, 2006). Some of these studies were completed strictly to measure the actual economic impact of a particular wind installation at the county level; while others were completed to estimate the economic impact of investing in a set amount of wind power within a state or region. All of the studies indicate that investment in wind power impacts rural economies by increasing jobs, income, and taxes. Furthermore, Pedden (2006) showed that, in communities with few other industries, the installation of wind farms can create a significant new industry that becomes a large percentage of the local tax base and contributes to local businesses.

At the same time, exceptionally small rural communities such as Sterling county also experience greater “leakage”¹⁵ of dollars out of their county even for those initial investments that occur locally. Counties such as Coke and Sterling typically have higher rates of leakage simply because, in a smaller more rural economy, there is less opportunity to invest project dollars in local goods and services. This translates into less indirect and induced impact from a wind installation than a larger community with the ability to provide a greater number of services. Indeed, as shown in Table 5, induced impacts for Horse Hollow during both construction and O&M were significantly higher than Capricorn Ridge: induced output during construction at Horse Hollow accounted for 14% of the total economic output versus 2% at Capricorn Ridge; during O&M, induced output at Horse Hollow was 18% of the total output versus 5% for Capricorn Ridge¹⁶. Thus, the combined Sterling and Coke Counties’ leakage rate is much greater than that of the combined Nolan and Taylor Counties region. Although the two study areas are approximately equal in geographic area, the Coke/Sterling area has a total employable population of only approximately 2450 while the Nolan/Taylor area has a total employable population of approximately 92,700. The Nolan/Taylor area also has a wind energy technician training facility and has actively engaged the wind industry in their community. Despite the differences between state and county economies, any level of local investment may be significant. Indeed even though the share of total project impact to these counties and the respective local areas is a fraction of the total project impact, the magnitude of the investment and resulting economic activity in these rural areas is relatively substantial.

Ultimately, the level of impacts and their distribution depends a great deal on the extent to which the locality of interest is able to directly participate in construction, operations, and ownership of a project. Because community wind projects (i.e., those involving some form of local ownership) are relatively rare, they constitute only about 2% of total wind capacity installed in the U.S. Therefore, impacts are more often a function of the ability of the local community to supply workforce, goods, and services both during construction and operations. In part, the ability of local businesses to participate in a project is a function of local economic development, developer preferences, and whether the industries affected by wind projects are situated in the community.

The Texas focus of this study provides a unique perspective on the question of economic impacts of wind energy. As noted earlier, Texas has over 10,000 MW of wind power and, if it were its own country, it would rank 6th in total installed capacity (Wiser and Bolinger, 2010). In addition, Texas has been noted for

its superior economic performance and employment growth, which has outpaced the U.S. by 1% on average for the last 40 years. Portions of Texas are quite rural but some areas have seen extensive (Giga Watt (GW) scale) wind energy deployment. Wind energy manufacturing has also begun to take hold with more than 15 announced or operating wind energy manufacturers in the state (Wiser and Bolinger, 2010). Texas represents a dynamic economy and one of the most mature wind energy markets in the country, but also demonstrates a great deal of diversity in structure and economic strength.

Over time, a variety of factors are likely to influence the total economic development activity captured by a state as well as the distribution of those economic impacts within a given state. As shown in Fig. 9, the emergence of a U.S. wind energy manufacturing sector has the potential to dramatically alter economic development activity from wind energy projects moving forward. The economic impacts from increased reliance on Texas-manufactured goods may not, however, generate a significant change in impacts for the counties where wind projects are sited. Instead, the development of wind manufacturing facilities is likely to greatly impact those communities immediately surrounding manufacturing facilities, but may not result in a significant impact beyond those localities.

Workforce development and technical training programs, (e.g., the Texas Wind Energy Institute at Texas Tech University) may also increase the number of skilled wind industry workers in Texas and in specific localities where large ongoing installations occur. However, if such programs are not accessible to individuals living in the localities where wind projects are sited, the development and training of skilled construction workers may not boost economic development at the level of the project host community. Economic development impacts to the state and a few specific localities where there is a skilled wind industry workforce will grow, but impacts to rural communities and counties may not grow and may even be reduced because other highly trained and specialized workers are available in neighboring counties or regions.

In today’s wind industry O&M workers often live in the communities where projects are sited; however, the development of remote monitoring and operations capabilities could reduce the demand for local labor. The continued maturation of the industry will undoubtedly have impacts on operations period economics; however, it is not clear what the impacts will be for those communities where wind projects are sited.

5. Conclusions

An analysis of two large wind energy projects constituting nearly 1398 MW of power capacity and completed in Texas in 2005–2008 are estimated to have resulted in 680 onsite construction period jobs and 4100 total construction period Texas jobs. During the operations period, statewide project impacts are estimated at 63 onsite jobs and 350 total jobs annually over the life of the project. Total statewide economic output from this level of wind energy development is estimated at \$1.8 billion (or \$1.32 million per MW of installed capacity) assuming a 20-year operating life for these installations. Impacts to the two county regions covered by each of these project footprints are estimated at 900 total jobs and \$110 million in total economic output during the construction period and 225 total jobs and \$31 million in annual economic output during operations. Onsite local jobs are estimated at 184 during construction and 63 during operations. The total economic impact to the local communities from the two wind farms is estimated at \$730 million over the assumed

¹⁵ Leakage is defined as the rate at which dollars invested in a specific project flow out of that area where the project is sited into the broader, state, national, or global economy.

¹⁶ The induced multiplier for Coke and Sterling county was approximately a factor of three less than that for Nolan and Taylor county, reflecting more limited opportunity for dollars to follow into the local service sector.

20-year life cycle of the farms, or \$0.52 million per MW of installed capacity.

Given the current level of impacts observed from wind projects and the potential for increased impacts via greater utilization of in-state manufacturing capacity and the development of trained wind industry specific laborers, Texas appears to be well positioned to see increasing impacts from continued wind development. The impact to rural communities and counties is also likely to be significant, especially those areas that have a larger resource base to draw from and the ability to provide a greater number of services. For very rural communities that have a more limited capacity to participate in wind energy projects, much of the economic impact of these projects is likely to accrue outside of the locality in which individual projects are built. In our research, the small community of Sterling City did see a greater leakage of dollars spent into nearby towns that provided more services. But by and large, these rural communities are realizing a significant amount of the potential economic impacts resulting from wind energy projects.

The phenomena observed here in Texas, where significant impacts accrue to states and where significant impacts to counties and communities are more variable and dependent on existing services, is likely to be consistent with the experiences of many other states and regions of the U.S. Moreover, typical policies seeking to increase the economic impacts of wind energy development to states (e.g., manufacturing investment incentives, workforce development programs) may be effective at increasing impacts to states but are less likely to greatly increase the ability of rural counties, where projects are sited, to capture increased economic impacts. Ensuring that rural communities where wind projects are built are able to capture more than O&M jobs, local property tax payments, and landowner lease payments is likely to require innovative mechanisms for inducing local community participation and partnership in new wind energy developments.

Acknowledgments

The work and contributions of TCU faculty and staff in this project were funded by NextEra Energy Resources. As specified under contract, TCU researchers had unrestricted access to company data, and complete independence in all aspects of the analysis, conclusions, and decision to publish the research. Participation in this research by the National Renewable Energy Laboratory was exclusively funded by the U.S. Department of Energy Wind and Water Power Program.

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